SOUND WAVES: CHARACTERISTICS AND APPLICATIONS

Waves: A Basic Introduction

✓ **What is a wave:**

- A **wave** is a disturbance that travels from one place to another and carries **energy**, without carrying the **matter** of the medium along with it.
- The particles of the material only vibrate (move to and fro) about their own fixed positions. They pass the disturbance on to the next particles.
- So, in a wave, **energy moves forward but the material stays at its place**.
- Waves are important because they carry energy and information from one place to another without sending any material. We see because of light waves, we hear because of sound waves and our mobile phones work because of radio waves.
- **Daily life examples of waves:**
  - **Ripples** (small waves on the surface of water) **on a pond:** when a small stone is dropped in a still pond, circular ripples move out and reach the edge. A dry leaf floating on the pond only moves **up and down** at its own place; it does not travel with the ripples. So only the disturbance (and its energy) travels, not the water.
  - **Mexican wave in a stadium:** each person only stands up and sits down at his or her own seat, but the 'wave' runs around the whole stadium.
  - **A rope tied to a pole:** when the free end is jerked up and down, a wave runs along the rope to the pole, but the rope does not move towards the pole.
  - **Sound** from a bell or a speaker, **light** from the Sun, **radio and mobile signals** and the shaking of the ground during an **earthquake** are all carried by waves.

✓ **How waves were discovered (history):**

- The idea of waves grew slowly over about 2,500 years, mostly from the study of **sound and music**. Only the most important steps are given here.

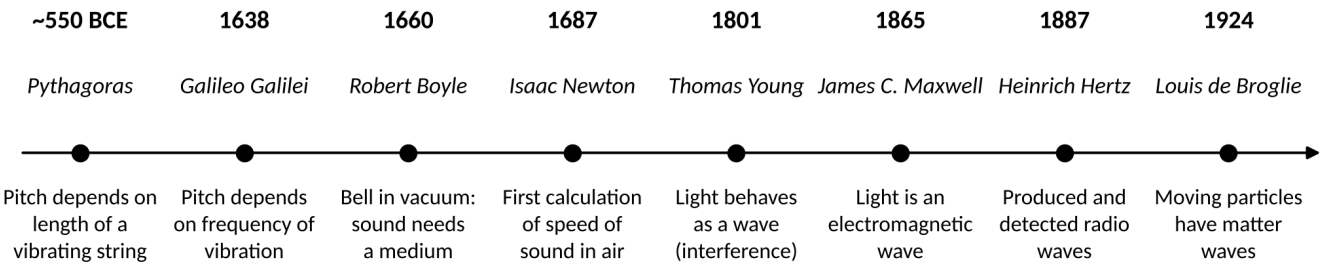


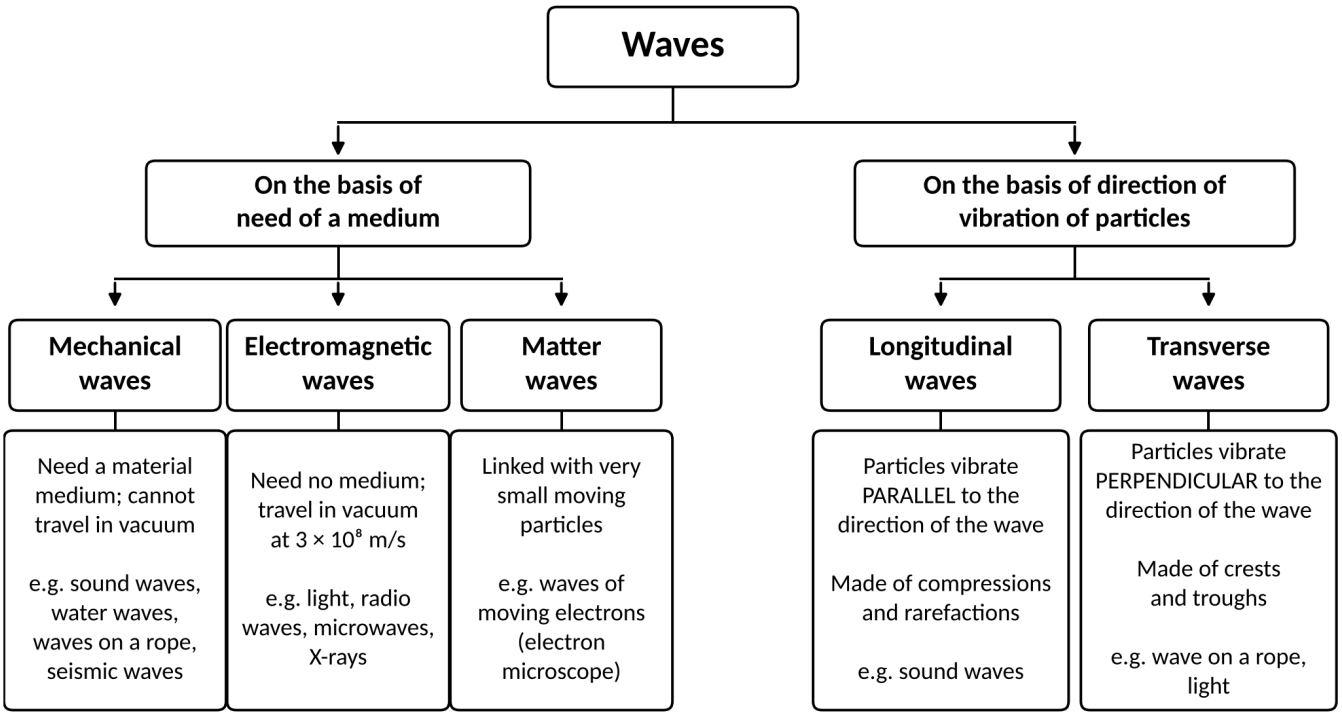
Fig. 1: Important steps in the discovery and understanding of waves

- **Pythagoras (Greece, about 550 BCE):** found that a shorter vibrating string gives a higher note. This was the first link between sound and vibration.
- **Galileo Galilei (Italy, 1638):** showed that the **pitch** of a sound depends on how many times the source vibrates in one second (frequency).

- **Robert Boyle (England, 1660):** showed with a bell in a jar emptied of air that **sound needs a medium** to travel.
- **Isaac Newton (England, 1687):** was the first to calculate the **speed of sound** in air.
- **Thomas Young (England, 1801):** proved that **light is a wave**.
- **James Clerk Maxwell (Scotland, 1865):** said that light is an **electromagnetic wave**.
- **Heinrich Hertz (Germany, 1887):** produced and detected **radio waves**. The unit of frequency, the **hertz (Hz)**, is named after him.
- **Louis de Broglie (France, 1924):** said that moving particles like electrons also behave as waves (**matter waves**).

✓ **Types of waves:**

- Waves are grouped in two ways: (A) on the basis of whether they need a medium, and (B) on the basis of the direction in which the particles vibrate.



Mechanical waves are of two types: longitudinal and transverse. Light (an electromagnetic wave) is a transverse wave.

Fig. 2: Classification of waves

- **(A) On the basis of need of a medium:**
  - **Mechanical waves:** waves that **need a material medium** (solid, liquid or gas) to travel. They cannot travel through vacuum.
    - **Examples:** sound waves, water waves, waves on a rope or a string, waves in a slinky and seismic (related to earthquakes) waves of an earthquake.
  - **Electromagnetic waves:** waves that **do not need any medium**. They can travel through vacuum also.
    - All of them travel with the same speed in vacuum, about 3,00,000 km per second (3×10^8 m s^{-1}).
    - Light from the Sun and the stars reaches the Earth through empty space because light is an

electromagnetic wave.

- **Examples:** radio waves, microwaves, infrared rays, visible light, ultraviolet rays, X-rays and gamma rays.

- **Matter waves:** waves linked with very small moving particles like electrons. They are used in the **electron microscope**, which shows very tiny objects like viruses.

(B) On the basis of direction of vibration of particles:

- **Mechanical waves are of two types:** longitudinal waves and transverse waves.
- **Longitudinal waves:** the particles of the medium vibrate **parallel** to the direction in which the wave travels (back and forth along the same line). The wave is made of **compressions and rarefactions**.
- **Transverse waves:** the particles of the medium vibrate **perpendicular** (at right angles (90°)) to the direction in which the wave travels. The wave is made of **crests and troughs**.
- **Note (light as a transverse wave):** light is also a transverse wave, but it has no particles to vibrate, as it can travel even in vacuum. In light, electric and magnetic effects (fields) change perpendicular to the direction of the wave. All electromagnetic waves are transverse.
- **Examples of longitudinal waves:** sound wave, a wave in a slinky pushed and pulled at one end.
- **Examples of transverse waves:** a wave on a rope moved up and down, light wave.

✓ **Graph of a transverse wave:**

- **The graph is drawn between the displacement of the particles (on the y-axis) and the distance along the direction of the wave (on the x-axis)**, at one instant of time.
- **Displacement** means how far a particle has moved up or down from its normal (rest) position.
- The dashed horizontal line is the **mean (rest) position**. It shows where the particles stay when there is no wave.

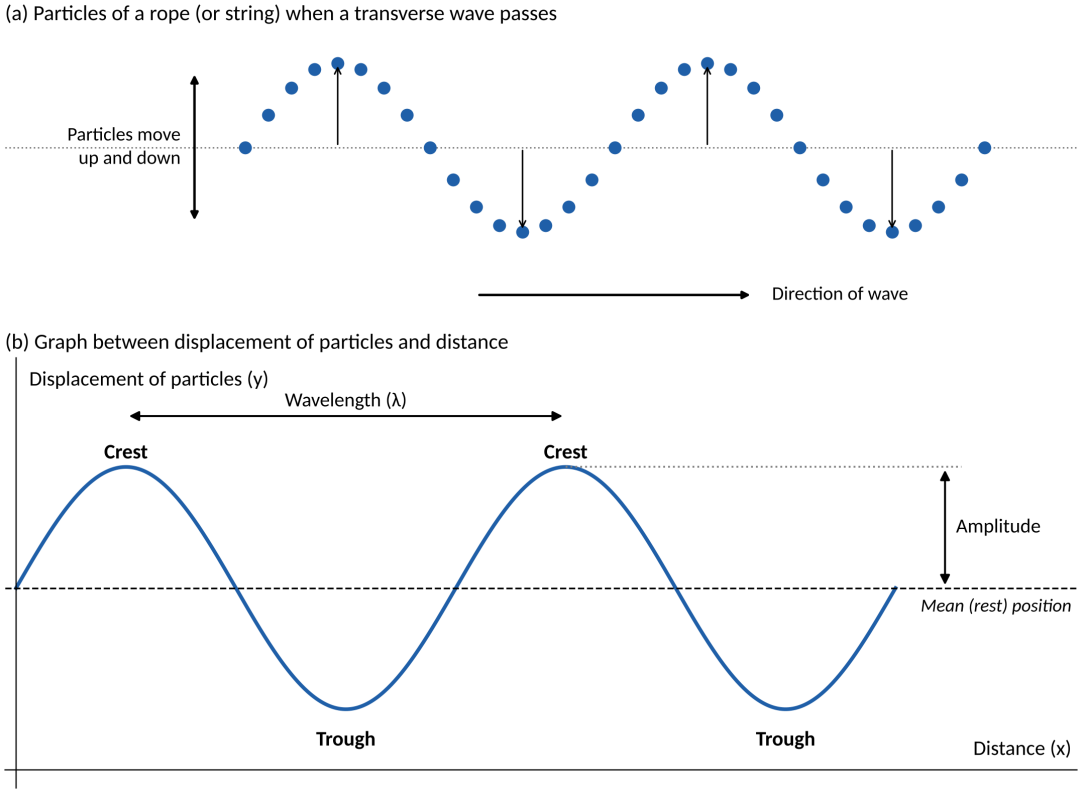


Fig. 3: Transverse wave: (a) particles of a rope moving up and down, (b) graph between displacement of particles and distance

- **Crest:** the highest point of the graph, where the particles have the maximum upward displacement.
- **Trough:** the lowest point of the graph, where the particles have the maximum downward displacement.
- **Wavelength (λ):** the distance between two consecutive (one after the other, without a gap) crests (or two consecutive troughs).
- **Amplitude:** the maximum displacement of the particles from the mean position, that is, the height of a crest (or the depth of a trough) from the dashed line.
- The shape of the graph looks just like the rope itself, as the particles really move up and down.

✓ **Graph of a longitudinal wave:**

- In a longitudinal wave, the particles do not move up and down. They move back and forth, so they **crowd together** at some places (compressions) and **spread out** at other places (rarefactions).
- So, for a longitudinal wave like sound, **the graph is drawn between the density of the medium (on the y-axis) and the distance from the source (on the x-axis)**, at one instant of time.
- **Density** means how closely the particles are packed in a region.
- The dashed horizontal line shows the **average density** (the density when there is no wave).

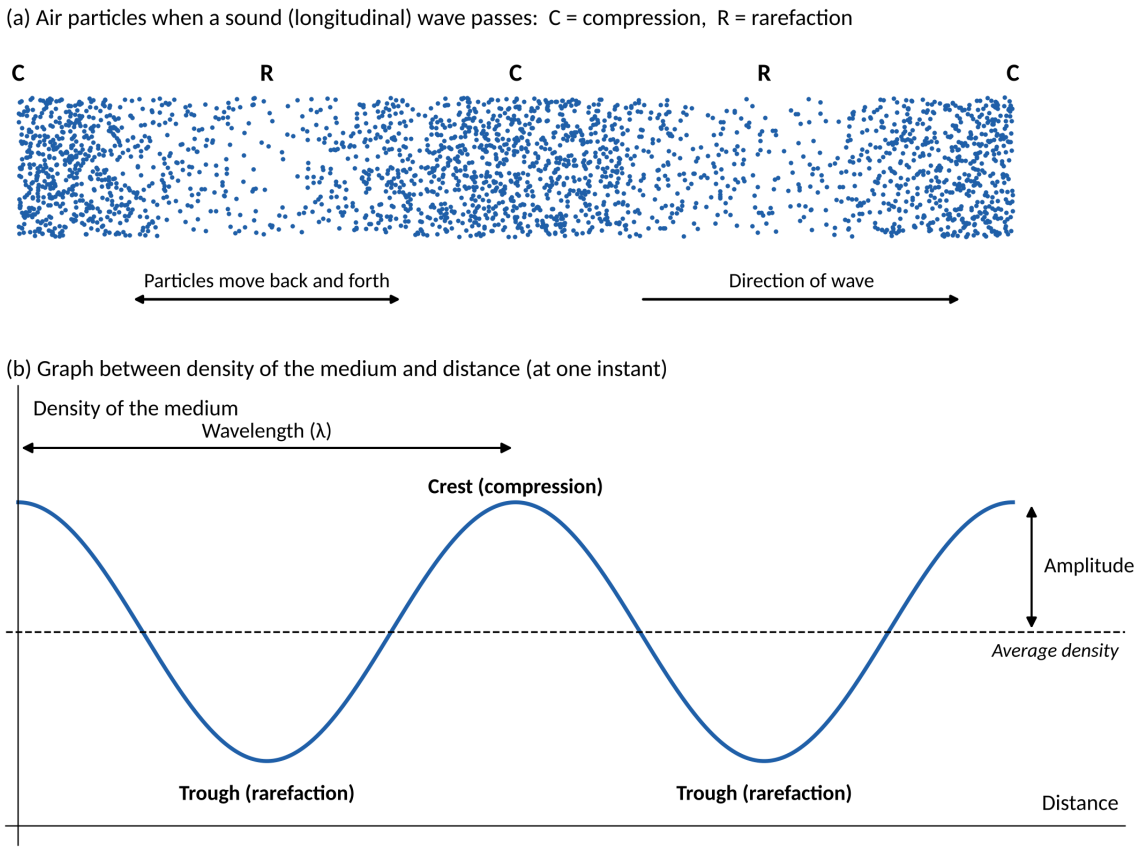


Fig. 4: Longitudinal wave: (a) compressions (C) and rarefactions (R) of air particles, (b) graph between density of the medium and distance

- **Compression (C)** has density more than the average, so it is shown as a **crest** (above the dashed line). The top of the crest is the maximum density.
- **Rarefaction (R)** has density less than the average, so it is shown as a **trough** (below the dashed line). The bottom of the trough is the minimum density.
- **Wavelength (λ):** the distance between two consecutive compressions (two consecutive crests) or two consecutive rarefactions (two consecutive troughs).

- **Amplitude:** the maximum change in density from the average density.
- The wave-shaped curve here does not show the shape of anything. The air particles do not move up and down; the curve only shows how the density changes from place to place.

Point	Graph of a transverse wave	Graph of a longitudinal wave
y-axis	Displacement of particles	Density of the medium
x-axis	Distance	Distance
Dashed line	Mean (rest) position	Average density
Crest	Maximum upward displacement	Compression (maximum density)
Trough	Maximum downward displacement	Rarefaction (minimum density)
Amplitude	Maximum displacement from the mean position	Maximum change in density from the average

- Sound is a **longitudinal mechanical wave**. All these terms are explained in detail for sound in the topics that follow.
- **Common doubts about waves cleared:**
  - **Water does not flow along with a wave:** a floating leaf or a paper boat only moves up and down at its place. Only energy moves forward.
  - **Light is much faster than sound:** light travels at about 3,00,000 km per second, while sound in air travels only about 340 m per second. That is why we see lightning first and hear thunder later.
  - **Sound from the Sun cannot reach us:** the space between the Sun and the Earth is almost empty (vacuum), and sound needs a medium. Light reaches us, as it does not need a medium.
  - **All waves are not the same:** all waves carry energy, but they differ in the need of a medium, in the direction of vibration and in speed.

Production of Sound

✓ **Sound in our life:**

- Sound is an everyday **sensory experience** (something we feel through our sense organs, here the ears). It makes us aware of our surroundings.
- Every day we hear many sounds: human voices, birds chirping, waves crashing on the seashore, leaves rustling, mobile phones ringing, vehicles honking, music and the claps of thunder.
- Sound is a **form of energy**. Energy can neither be created nor be destroyed. It only changes from one form to another.
- So, some other form of energy gets changed into sound energy. The **energy of a vibrating object** (its mechanical energy) gets changed into sound energy.
- In this chapter we learn how sound is produced and how it reaches our ears.
- **Example:** the muscular energy used to pluck a string or strike a drum makes it vibrate, and this vibration produces sound.
- **Two interesting examples:**
  - **Example 1:** Two astronauts repairing the arm of a space station during a spacewalk cannot talk to each other or hear the sound of metal clanking the way they do on the Earth. This is explained in the topic on propagation (travelling or spreading from one place to another) of sound.
  - **Example 2:** Most bats find their prey in the dark at night by using sound (ultrasonic waves and

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their echoes). This is explained in the topic on echolocation.

✓ **Ways to produce sound:**

- In arts lessons of earlier grades, we learnt a few ways to produce sound, such as playing the bansuri (flute) and other musical instruments.
- **Sonority:** some metals produce a ringing sound when struck with an object. This property is called **sonority**. Example: the **taal** (a pair of small metal cymbals).



Fig. 5: Taal, a musical instrument made of metal

- **Sound is produced by vibration:**
  - **Sound is produced by vibrations**. As long as an object vibrates, it produces sound; once the vibration stops, the sound also stops (see Activity 1).
  - **Vibration** is the periodic (repeating again and again after equal intervals of time) to and fro motion (**oscillations**) of an object.
  - The object that produces sound is called the **source** of the sound.
  - Sound can be produced by vibrating **strings**, **membranes** (a thin, flexible sheet), **air columns** and many other vibrating objects.
  - In most musical instruments, **more than one vibrating part** is involved in producing sound.
  - **Example 1:** a stretched rubber band produces sound only while it is vibrating.
  - **Example 2:** pulling (plucking) a stretched string or striking a metal object makes it vibrate, and this produces sound.
  - **Example 3:** when we blow through a **bansuri** (flute), the **air inside the hollow pipe** vibrates and produces sound.
  - **Example 4:** in a sitar (a plucked Indian string instrument with a long neck), the strings vibrate and the hollow wooden body also vibrates with them, which makes the sound louder.

Way of producing sound	Vibrating part	Examples
Plucking or bowing a stretched string	String	Sitar, veena, guitar, sarangi, ektara, tanpura
Striking a stretched membrane	Membrane (skin)	Tabla, dholak, mridangam, drum
Blowing air into a pipe	Air column	Bansuri (flute), shehnai, whistle, harmonium reeds
Striking a metal or solid object	Whole object	Taal, bell, ghungroo, jal tarang, tuning fork

✓ **How humans and animals produce sound:**

- While talking or singing, if we gently touch our throat, we feel **vibrations**.
- In humans and some other animals, sound is produced by the vibration of the **vocal cords**.
- **Vocal cords** are tightly stretched **muscular flaps** (thin folds made of muscle) located inside the **voice box** or **larynx**, in the throat.

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- When we speak, air from the lungs passes between the vocal cords and makes them vibrate. When we are silent and breathing, they stay open.
- The **tongue, lips, mouth and nasal cavity** (the hollow space inside the nose) in humans help in changing the sound into **speech or music**.

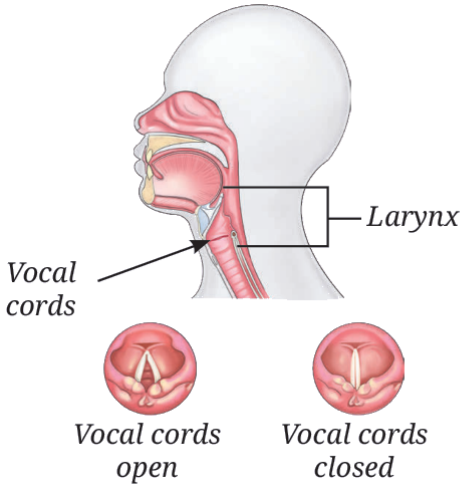


Fig. 6: Vocal cords inside the larynx (open and closed)

- Some animals produce sound by **striking or rubbing** certain body parts. Example: **grasshoppers and crickets** rub their wings or legs to produce sound.

✓ **Tuning fork:**

- A **tuning fork** is an instrument often used for experiments with sound.
- It is a **U-shaped metal bar with a stem**. It is usually made of **steel or aluminium**.
- The two sides of the 'U' are called **prongs** or **tines**.
- The prongs are struck on a **rubber pad** to make them vibrate. It should never be struck on a hard surface, as that can damage it and change its sound.
- A vibrating tuning fork produces sound. When one of its vibrating prongs touches water, small waves form on the water. This shows that the prongs are really vibrating (see Activity 2).
- A vibrating tuning fork can be heard when it is held near the ear in **different orientations** (directions in which something is turned or held). This shows that sound spreads in many directions from the source.

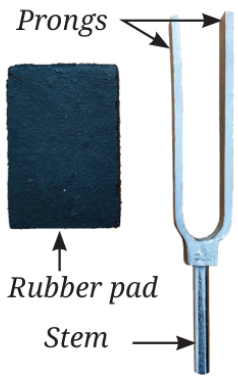


Fig. 7: A tuning fork (prongs and stem) and a rubber pad

- **Kongthong: the Whistling Village:**
  - **Kongthong** is a village near **Shillong in Meghalaya**. It is also known as the **Whistling Village**.

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- Here, every person has a **'tune name'** that can be sung or whistled, in place of (or along with) a word name.
- This unique tradition is called **Jingrwai Iawbei**.
- It begins at birth, when a **mother composes a lullaby-like tune** for her child. Villagers call each other using these tunes, even across the forest.

Propagation of Sound

✓ **Sound travels through solids, liquids and gases:**

- Sound is produced by vibrating objects. It reaches our ear from the source through some material.
- Sound can travel or **propagate** (travel or spread from one place to another) through **solids, liquids and gases**.
- The material through which sound propagates is called a **medium**. Sound propagates from its source to us through a medium.
- A space where there is **no medium (no matter)** is called a **vacuum**.
- **Example 1 (solid):** when a friend gently knocks or scratches a desk, the sound is heard clearly with the ear placed against the desk. So **sound travels through solids** (see Activity 3).
- **Example 2 (liquid):** when two metal spoons are tapped together under water, the sound is still heard. The sound reaches us after travelling through water and air. If sound did not travel through liquids, we would not have heard it. So **sound travels through liquids** (see Activity 4).
- **Example 3 (gas):** we hear each other speak through air every day. So **sound travels through gases**.

✓ **Sound needs a medium to propagate:**

- **Vacuum bell jar experiment:**
  - An **electric bell** is kept inside a glass **bell jar**. The bell is switched on and the loudness of its sound is noted.
  - Air is slowly sucked out of the bell jar using a **vacuum pump**. The sound becomes **fainter** (weaker).
  - When a near vacuum is reached, **almost no sound** is heard, even though the bell can be **seen ringing**.
  - When air is let back into the jar, the sound is heard again and it slowly becomes as loud as before.

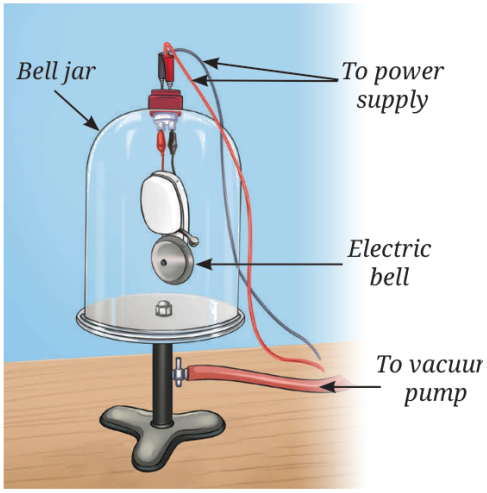


Fig. 8: Vacuum bell jar experiment

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- **Conclusion:** sound **cannot propagate in vacuum**. Sound needs a **medium** to propagate. The medium can be a solid, a liquid or a gas.
- We can still see the bell ringing, as light does not need a medium. But we cannot hear it, as sound needs a medium.
- **Note (why 'almost no sound' and not 'no sound'):** a pump cannot remove all the air, so a little air is always left. Also, the bell is fixed to the jar through its stand and wires, and some sound passes through these solids. So a very faint sound may still be heard.

Astronauts in space:

- Outer space is a **near vacuum**. So sound cannot propagate there.
- Astronauts in spacesuits doing spacewalks **cannot directly hear each other speak** or hear sounds like two metal objects clanking together.
- They talk to each other through **special devices** (radio sets) fitted in their spacesuits, as radio waves can travel through vacuum.
- If the two astronauts touch their helmets together, they can hear each other faintly, as sound then travels through the solid helmets.

Sound Waves

✓ **Sound spreads in many directions:**

- Sound propagates in **many directions** from a source.
- The directions in which sound propagates can depend on the **shape of the source**.
- For simplicity, sound is usually shown as moving in only **one direction**.
- **Example:** a vibrating tuning fork can be heard when held near the ear in different orientations.

✓ **Slinky as a model of the medium:**

- A **slinky** is a long, flexible metal or plastic spring toy. It is used as an **analogy** (a comparison with something familiar to explain an idea) for the medium through which sound travels.
- Vibrating one end of the slinky (pushing and pulling it) is like a source producing sound (see Activity 5).
- Regions are formed where the turns of the slinky are **closer together** than usual, and other regions where they are **more spread out**.
- These closely spaced and spread out regions **travel along the slinky**.
- But a mark made on one turn **does not travel** along the slinky. It only **oscillates** (move to and fro again and again about a fixed position) **about its position of rest**, **parallel** to the direction of the disturbance.
- Each turn oscillates about its own position, while the **disturbance** travels from one end of the slinky to the other.
- Sound moves through a medium in the same way. The closeness or spreading out of the turns of the slinky stands for the **higher or lower density of air** through which sound travels.

✓ **Piston model: compressions and rarefactions:**

- Take a long tube filled with air. It has a **piston** (a tight-fitting disc that can slide) at one end and is open at the other end. The piston can oscillate back and forth inside the tube.
- The oscillating piston is a **simple and idealistic** (simplified, perfect form that is easy to imagine) **model** of the source of sound. It helps us understand how sound travels through air.

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- **(a) Piston not oscillating:** the air inside the tube has a certain **uniform density**, called the **average density**. (The dots in the figure stand for air particles.)
- **(b) Piston moves forward:** it pushes and displaces (pushes something away from its place) the nearby air particles forward. This **compresses** the air and increases its density in a small region close to the piston. A region of air with **higher density** (compared to the average density) is called a **compression (C)**.
- The compressed air particles **collide** (hit or bump into each other) with the particles ahead and pass the compression forward. These particles collide with their neighbours, and so on. So the high-density compression **moves forward** through the air, but the air particles themselves **do not travel** with it.
- **(c) Piston moves backward:** the air particles move backward towards the piston, and the air in a small region close to the piston becomes **less dense**. A region of air with **lower density** (compared to the average density) is called a **rarefaction (R)**.
- Due to collisions with nearby particles, the rarefaction also moves ahead, while the particles only oscillate about their **mean positions** (normal resting positions).
- **(d) Piston oscillates continuously:** compressions and rarefactions are produced **one after the other** (alternately). They travel away from the source as a series of compressions and rarefactions.

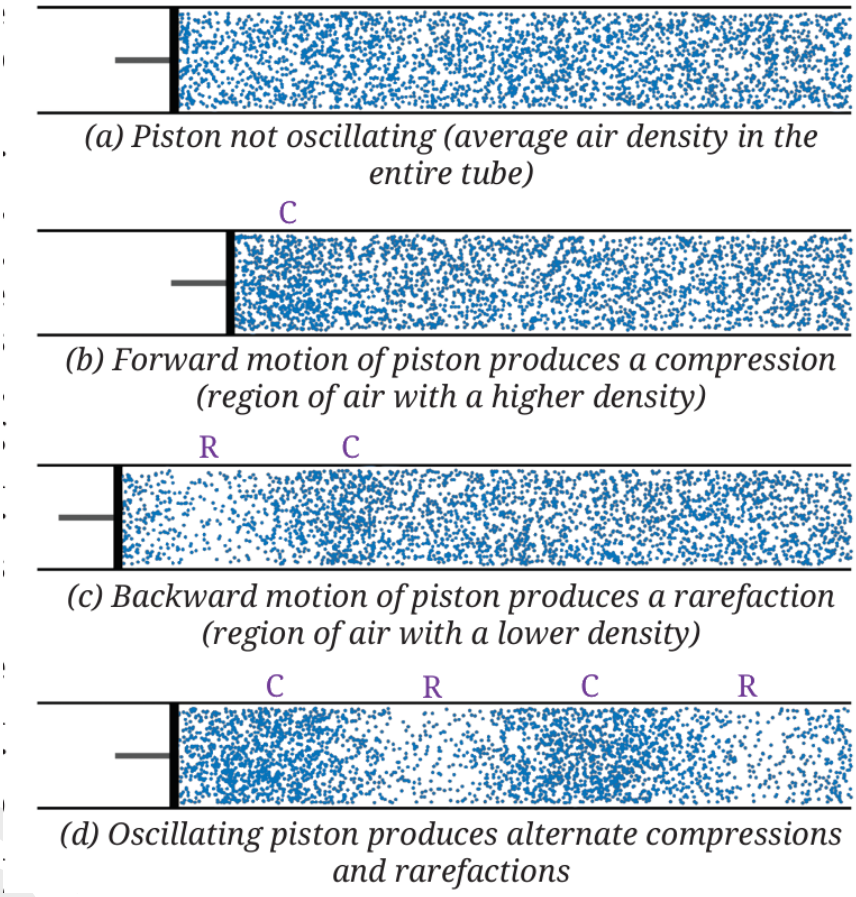


Fig. 9: Density of air in a tube with an oscillating piston: (a) no oscillation, (b) compression, (c) rarefaction, (d) alternate compressions and rarefactions

- **Sound wave:**
  - The disturbance made of a **series of alternating compressions and rarefactions** moving through a medium, **without the actual flow of the particles of the medium**, is called a **sound wave**.

- The direction in which the wave travels is called the **direction of propagation** of the wave.
- The particles of the medium **do not travel with the wave**. They only vibrate about their mean positions.

Spherical waves:

- When the medium is not closed inside a tube, its vibrating particles collide with the surrounding particles in **all directions**. So the sound wave spreads out in all directions.
- A small source producing sound continuously sends compressions and rarefactions in all directions through the surrounding medium. These spread as **spherical** (shaped like a ball (sphere)) **waves** (like an expanding ball).
- When these waves reach a listener, they are **perceived** (sensed and understood by the brain) as sound.

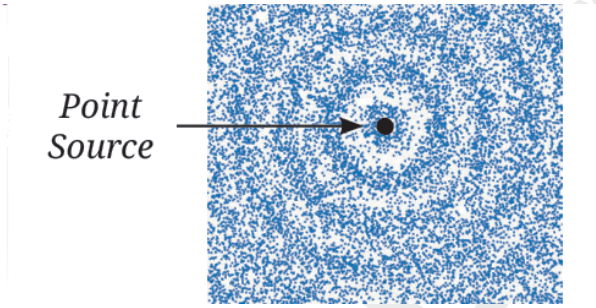


Fig. 10: Sound from a point source spreads in all directions as spherical waves

✓ **Sudden loud sounds: firecrackers, thunder and sonic boom:**

- Sudden loud sounds, like the explosion of firecrackers or a clap of thunder, are produced in a slightly different way.
- When air or gases are **heated very quickly**, they **expand very quickly** in a short time.
- This rapid expansion creates a **sudden disturbance in the air density** that travels outward.
- When this disturbance reaches our ears, we hear it as a **loud sound pulse**.
- A similar but more complex disturbance is produced when a **supersonic** (faster than the speed of sound) **aircraft** flies at a speed **higher than the speed of sound**. It creates a very loud sound called a **sonic boom**.



Fig. 11: A supersonic aircraft

✓ **Longitudinal waves:**

- In the piston model, the air particles move (are displaced) **parallel** to the direction in which the wave travels.
- In sound waves, the particles of the medium vibrate **back and forth parallel to the direction of**

propagation of the disturbance.

- Waves in which the particles vibrate in a direction **parallel to the direction of wave propagation** are called **longitudinal waves**.

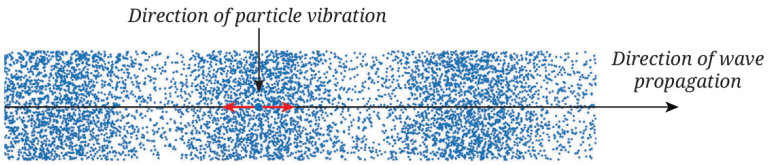


Fig. 12: A longitudinal wave: particles vibrate parallel to the direction of wave propagation

✓ **Mechanical waves:**

- If there is no medium (no particles), sound waves **cannot** propagate, because there is nothing to vibrate and pass on the disturbance.
- Waves that need a **material medium** for propagation are called **mechanical waves**.
- Sound wave is a type of mechanical wave.
- Mechanical waves are of two types: **longitudinal waves** and **transverse waves**.

✓ **Transverse waves:**

- Sound wave is an example of a longitudinal wave, where the particles vibrate parallel to the direction of the wave.
- In a **transverse wave**, the particles vibrate in a direction **perpendicular to the direction of wave propagation**.

Longitudinal wave	Transverse wave
Particles vibrate parallel to the direction of the wave.	Particles vibrate perpendicular to the direction of the wave.
Made of compressions and rarefactions.	Made of crests and troughs.
Example: sound wave, push-pull wave in a slinky.	**Example:** wave on a rope, light wave.

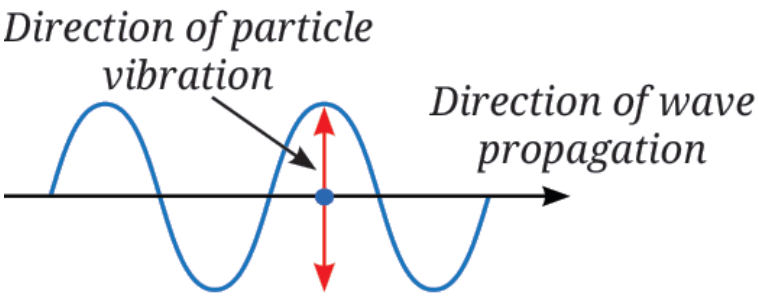


Fig. 13: A transverse wave: particles vibrate perpendicular to the direction of wave propagation

- **Seismic waves:**
  - **Earthquakes** produce **seismic waves** that travel through the Earth.
  - These waves can be **longitudinal or transverse**.
  - The **longitudinal seismic waves** are the **first to be detected** by **seismographs** (instruments that detect earthquakes), as they travel faster than the transverse ones.
- **Not all waves are mechanical:**

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- Not all waves need a medium to propagate.
- **Light** is a **transverse wave** that can travel through **vacuum**. This is why light from the Sun and other stars can reach the Earth.
- Waves will be studied in more detail in higher grades.

Energy of Sound Waves

➢ **Sound is a form of energy**.

➢ When the source of sound vibrates, it **transfers energy** to the surrounding medium.

➢ As sound waves move in a medium, the vibration of the particles and their **collisions** with other particles **transfer the energy** forward.

➢ In the propagation of a sound wave, it is the **energy** that is transferred, **not the particles** of the medium.

➢ **Example:** when a loud sound is produced near a container covered with a tightly stretched sheet, grains sprinkled on the sheet **jump**, even though the source does not touch the sheet (see Activity 6). Sound travels through air, reaches the sheet and makes it **vibrate**, and this vibration makes the grains move.

- **Particles are never at rest:**
  - Particles in a medium are never truly at rest. They are always **randomly vibrating** due to **thermal energy** (heat energy).
  - When a sound wave passes through the medium, it **temporarily increases** the vibration of these particles.
  - After the wave passes, the particles return to their usual random motion.

✓ **Microphone and speaker:**

- A **microphone** captures sound from various sources. It changes **sound energy into electrical energy**.
- When we speak or sing into a microphone, the sound waves make a thin **membrane**, called a **diaphragm** (a thin sheet that can vibrate), vibrate. These vibrations are changed into an **electrical signal**.
- A **speaker** does the opposite. An electrical signal makes a **cone or diaphragm** fitted inside the speaker vibrate, and this produces sound.
- If all parts work properly, the sound from the speaker closely matches the sound that was first captured.

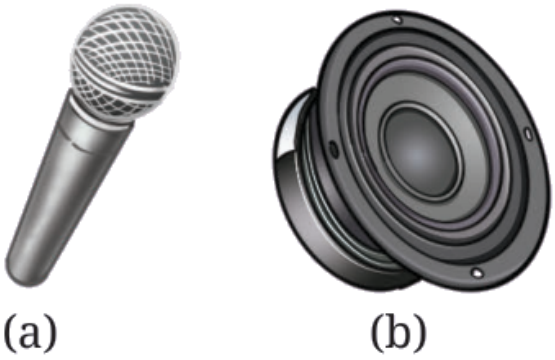


Fig. 14: (a) A microphone and (b) a speaker

Graphical Representation of a Sound Wave

➢ As a sound wave moves, at any given instant of time, the **density of the medium** changes periodically

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with **distance** from the source.

- This can be shown on a graph with **distance on the x-axis** and the **density of the medium on the y-axis**.
- The **average density** is marked as a horizontal dashed line.
- At a given instant, the density goes **above and below** the average density as the distance changes.

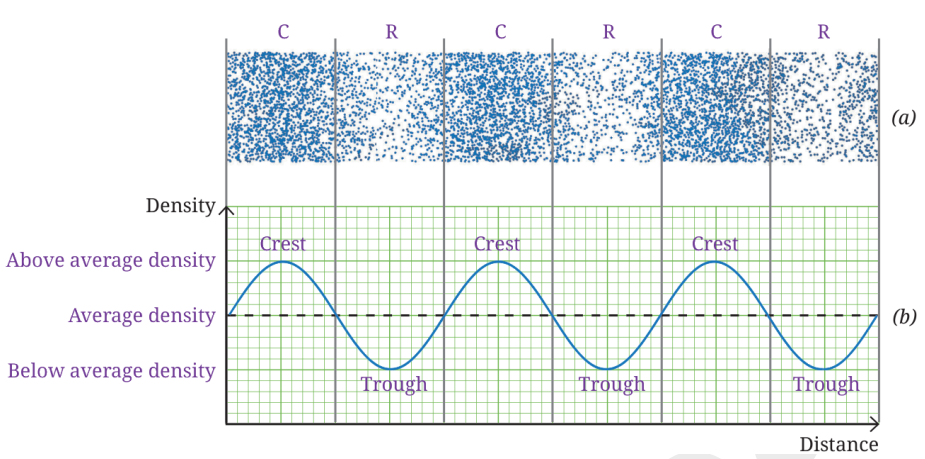


Fig. 15: For a sound wave: (a) variation of density of the medium, (b) graph of density against distance

- In a **compression**, the density rises **above** the average density. The highest point stands for the **maximum density**.
- After that, the density decreases and falls **below** the average density in the **rarefaction**. The lowest point stands for the **minimum density**.
- The highest point of the graph is called the **crest** and the lowest point is called the **trough** of the wave.
- So, on the graph, a compression is shown by a crest and a rarefaction by a trough.
- Another way to draw a sound wave is to plot the change in density of the medium with **time**, at one particular place in the medium.
- In this **density–time graph**, time is on the x-axis and the density at that fixed point is on the y-axis. The time between two consecutive crests on this graph is the **time period (T)** of the wave.

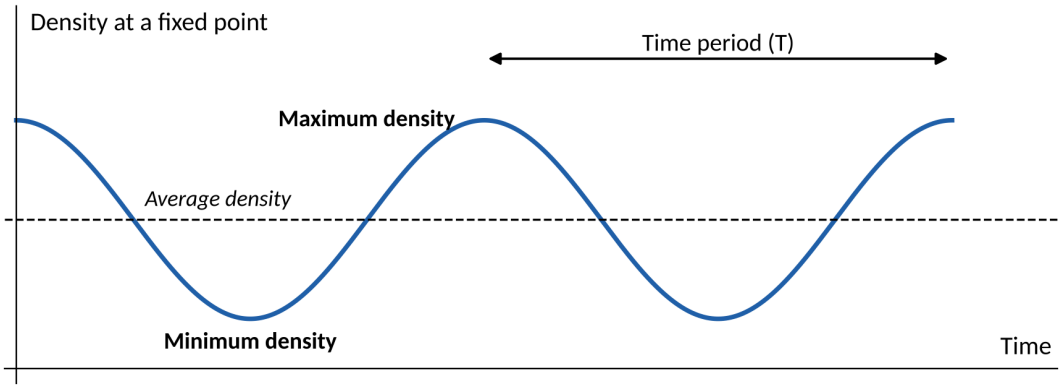


Fig. 16: Graph between density and time at a fixed point in the medium

Characteristics of a Sound Wave

✓ **Wavelength, frequency and time period:**

Wavelength:

- The distance between **two consecutive crests** or **two consecutive troughs** is called the **wavelength** of a wave.

- In terms of the medium, it is also the distance between two consecutive compressions or two consecutive rarefactions.
- Wavelength is shown by **λ** (Greek letter **lambda**). Its SI unit is the **metre (m)**.

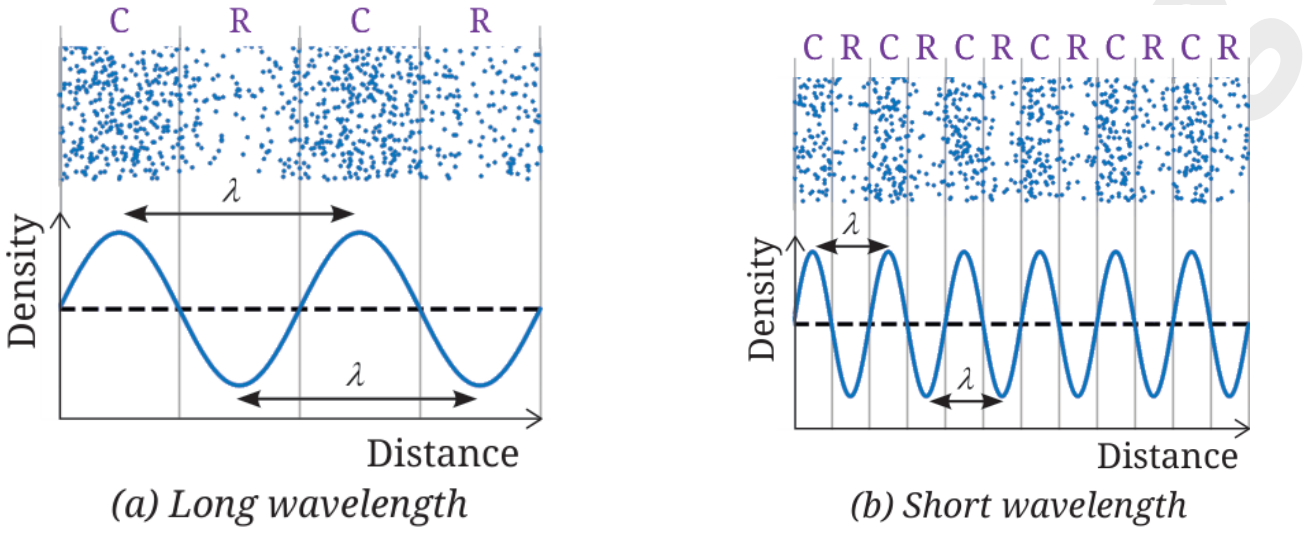


Fig. 17: Sound waves with (a) long wavelength and (b) short wavelength

One complete oscillation:

- When a sound wave passes a fixed position in the medium, the density at that position changes again and again between a maximum density (crest) and a minimum density (trough).
- The change in density at a fixed point from **maximum to minimum and then again to maximum** (or from minimum to maximum and back to minimum) makes **one complete oscillation**.

Frequency:

- The **number of density oscillations** (repeated to and fro movements about a fixed position) **at a fixed point per unit time** is the **frequency** of the sound wave.
- It tells us how often the density changes happen at a given position.
- Frequency is shown by **ν** (Greek letter **nu**).
- Its SI unit is **per second (s^{-1})**, also called a **hertz (symbol Hz)**. 1 kHz (kilohertz) = 1000 Hz.

Time period:

- The **time taken for one complete density oscillation** at a fixed point is the **time period** of the wave.
- It is shown by the symbol **T**. Its SI unit is the **second (s)**.
- Time period and frequency are **inversely** (in the opposite way (when one increases, the other decreases)) **related**: a shorter time period means a higher frequency.
- **ν = 1/T** or **T = 1/ν** ..... (Equation 1)
- **Example:** if a source makes 100 oscillations in 1 second, each oscillation takes 1/100 s = 0.01 s.

Single frequency sounds:

- Most everyday sounds contain a **mixture of many frequencies**.
- Nearly **single frequency** sounds can be made by striking a **tuning fork** or by **whistling** with the mouth.
- Free mobile apps, such as **Phyphox**, can be used to produce and identify sounds of different frequencies. Using its 'Audio Scope' and whistling carefully often gives a smooth regular graph like

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the one shown for wavelength. This shows that a whistle is almost a single frequency sound.

- **Frequency of musical notes:**
  - When the musical notes **Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa** are sung one after another, the frequency is **lowest for 'Sa'** and gradually **increases** for the other notes (see Activity 7).
  - Each musical note has a **distinct frequency**, which makes it sound different.
  - The upper (last) 'Sa' has about **double** the frequency of the first 'Sa'.
  - The same app can be used to compare the frequencies of other sounds, including the voices of friends.
- **Solved Example 1:**
  - **Question:** There are 10 density oscillations in 2 seconds at a given position. Find (i) the frequency of the sound wave and (ii) its time period.
  - (i) Frequency = number of oscillations ÷ time taken = 10 ÷ 2 s = **5 Hz**
  - (ii) Time period = time for one oscillation = 2 s ÷ 10 = **0.2 s**
  - Check: v = 1/T = 1/0.2 s = 5 Hz.

✓ **Amplitude and intensity of sound waves:**

- **Amplitude:**
  - Sound moves as density oscillations of the medium through compressions and rarefactions.
  - The **amplitude** of a sound wave is the **maximum change in the density of air** in a compression (or a rarefaction) **compared to the average density**.
  - A larger change in density means a **larger amplitude**. On the graph, it is the height of the crest (or depth of the trough) from the average density line.

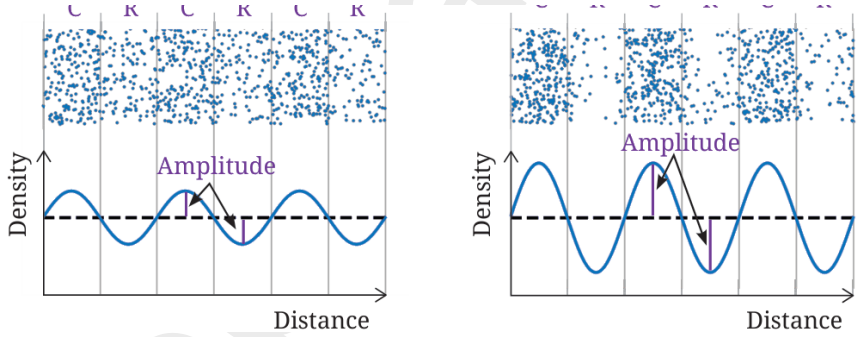


Fig. 18: Sound waves with (a) low amplitude and (b) high amplitude

- **Amplitude and energy:**
  - The amplitude and the amount of **energy** a wave carries are related.
  - A wave with a **larger amplitude carries more energy** than a wave with a smaller amplitude.
  - This can be seen in Activity 6. When the plate is struck harder, more energy goes to the particles of the surrounding medium and they are displaced more from their mean positions. So the sheet vibrates more and the grains **jump higher**.
- **Intensity:**
  - The amount of sound energy passing through a **unit area** held **perpendicular** to the direction of propagation of the sound wave in **unit time** is called the **intensity of sound**.
  - As a sound wave travels away from its source, it **spreads over a larger area**.
  - Energy must be **conserved** (it cannot be lost or created), so the same amount of energy is now

Conceptual Learning

spread over a larger area.

- Hence, the **intensity decreases with distance** from the source. That is why sound becomes fainter as we move away from it.
- Sounds produced with a larger amplitude at the start carry more energy, so they can travel a larger distance before their intensity becomes zero.

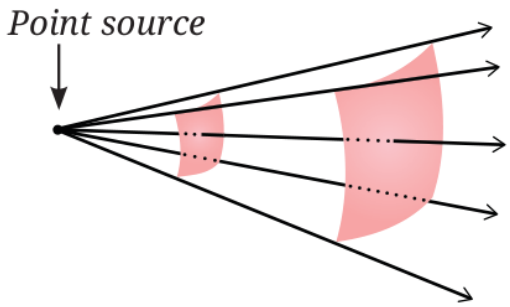


Fig. 19: Sound energy spreads over a larger area as the distance from the source increases

✓ **Speed of sound:**

- The **speed of sound** tells us how fast sound waves move through a medium. In terms of compressions and rarefactions, it is how fast these density disturbances move through the medium.
- The speed of sound is the **distance travelled by a point on a wave**, such as a crest (or a trough), in **unit time**.
- **Relation between speed, wavelength and frequency:**
  - For a sound wave of given frequency, the distance between two consecutive crests is one wavelength (λ).
  - This distance is covered by the disturbance in one time period (T).
  - Speed = distance ÷ time, so **$v = \lambda/T$**
  - As v = 1/T (Equation 1), we get **$v = \lambda \times v$** ..... (Equation 2)
  - In words: **speed = wavelength × frequency**
- **Speed of sound in different media:**
  - The speed of sound **depends on the medium** through which it travels.
  - It travels **fastest in solids**, **slower in liquids** and **slowest in gases**.
  - The particles in solids are packed very close and are held by strong forces, so a push given to one particle is passed on to the next very quickly. In gases, the particles are far apart and weakly held, so it is passed on slowly.
  - Sound travels about **4 to 5 times faster in water** and usually **15 to 20 times faster in solids** than in air.

Table 1: Speed of sound in different media at 15 °C (at atmospheric pressure)

State	Substance/Medium	Approximate speed
Solid	Steel	5000 m s^{-1}
Liquid	Water	1500 m s^{-1}
Gas	Air	340 m s^{-1}

- **Effect of temperature and humidity:**

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- The speed of sound in air also depends on the **temperature** and **humidity** (amount of water vapour in the air).
- When the temperature or humidity increases, the **speed of sound increases**.
- **Data:** the speed of sound in dry air is about **331 m s^{-1} at 0 °C** and nearly **344 m s^{-1} at 22 °C**.

Speed does not depend on frequency:

- In most media, such as air, the speed of sound depends **only on the medium**, and **not on the source or the frequency**.
- If the frequency of the source changes, the **wavelength** of the sound wave in the medium changes, but the **speed remains the same**. (As $v = \lambda\nu$, if ν becomes double, λ becomes half.)
- Some special materials, such as **porous** (having many tiny holes) **foams** or specially engineered structures, can behave differently.
- **Note (what decides what):** the **frequency** is decided by the **source** and does not change when sound goes from one medium to another. The **speed** is decided by the **medium**. So, when sound enters a new medium, its **wavelength changes** ($\lambda = v/\nu$).

Examples on speed of sound:

- **Example 1 (what if the speed of sound depended on frequency):** when a singer performs with instruments, the high notes and low notes produced together would reach the listener at different times. The music would get mixed up and would not sound pleasant. We enjoy music only because all frequencies travel with the same speed in air.
- **Example 2 (sound through air and through a steel fence):** Gunjan places her ear on a steel fence, and her friend 340 m away knocks the fence with a metal object. Gunjan hears the knock twice: first through the steel and then through the air, as sound is much faster in steel. Two sounds are heard separately only if they reach the ear at least 0.1 s apart; here the gap is about 0.93 s, so she hears both separately. (Full working is given in the solved questions.)

Solved Example 2:

- **Question:** Human hearing roughly spans 20 Hz to 20 kHz. Find the wavelengths in air for these two frequencies. (Speed of sound in air = 344 m s^{-1})
- Speed = frequency × wavelength, so wavelength (λ) = speed ÷ frequency
- (i) For ν = 20 Hz: λ = 344 ÷ 20 = **17.2 m**
- (ii) For ν = 20 kHz = 20000 Hz: λ = 344 ÷ 20000 = 0.0172 m = **1.72 cm**
- So a low frequency sound has a long wavelength and a high frequency sound has a short wavelength.

Solved Example 3:

- **Question:** During a thunderstorm, the time gap between seeing the lightning flash and hearing the thunder is 5 s. Estimate the distance of the lightning strike. (Speed of sound in air = 340 m s^{-1}; light, with speed 3,00,000 km s^{-1}, reaches almost instantaneously.)
- Lightning is seen before thunder is heard, as sound travels much slower than light.
- Distance = $v \times t$ = 340 m s^{-1} × 5 s = **1700 m**
- So the lightning struck about **1.7 km** away.

Solved Example 4:

- **Question:** From the graph of a sound wave travelling in steel (given below), find its wavelength. Find its frequency and time period, if the speed of sound in steel is 5000 m s^{-1}.

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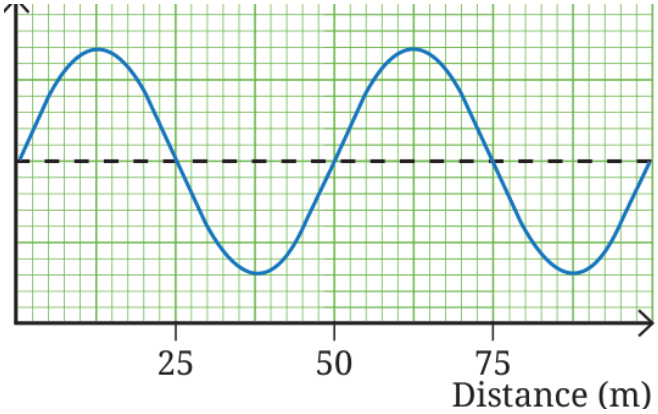


Fig. 20: Graph of a sound wave travelling in steel (density against distance in metres)

- From the graph, two consecutive crests are 50 m apart. So wavelength **λ = 50 m**.
- Using Equation 2, frequency v = v/λ = 5000 ÷ 50 = **100 Hz**
- Using Equation 1, time period T = 1/v = 1/100 = **0.01 s**

✓ **Human perception of sound:**

- The **physical properties** of sound, such as time period, wavelength, frequency, amplitude and speed, are well-defined and can be **measured**.
- But how we **experience** sound is **subjective** (depending on the person who hears or feels it).
- Human perception of sound is described by terms such as **loudness** and **pitch**. These do not have simple relations with the physical properties.
- **Pitch:**
  - How **frequency is perceived** by humans is called **pitch**.
  - In general, high pitch sounds have **higher frequency** and low pitch sounds have **lower frequency**, although the exact mathematical relation is complicated.
  - Sounds felt as **shrill** (high and sharp) have a **high pitch**, and **deep** sounds have a **low pitch**.
  - **Note (pitch is not loudness):** pitch depends on **frequency**, while loudness depends on **amplitude**. A high pitch sound can be soft and a low pitch sound can be loud.
  - **Examples of high pitch:** a whistle, a siren, a mosquito's buzz (high pitch but soft).
  - **Examples of low pitch:** thunder, the rumble (deep and continuous sound) of an aircraft, a lion's roar (low pitch but loud).
- **Why our voices sound different:**
  - Everyone's voice sounds unique. We can often recognise our teacher or friends at once when they call our name.
  - Male, female and children's voices differ not only in their **frequency** but also in how the sound is **shaped by the throat, mouth and nasal cavities**.
  - During **adolescence** (the teenage years, when a child grows into an adult), the vocal cords of boys become **longer and thicker**. So they vibrate **less frequently** (lower frequency), and this 'deepens' their voice.
- **Audible range:**
  - Humans can hear only a **limited range** of sound frequencies (see Activity 8).
  - The **audible range** or human hearing range is from **20 Hz to 20,000 Hz (20 kHz)**.
  - This range is different for different people and it **decreases with age**. Older people often cannot

Conceptual Learning

hear very high frequencies.

- Sound waves with frequency **below 20 Hz** are called **infrasonic waves** (infrasound).
- Sound waves with frequency **above 20 kHz** are called **ultrasonic waves** (ultrasound).
- Humans cannot hear infrasound or ultrasound, but some animals can.
- **Dogs, cats, bats and dolphins** can detect **ultrasound**. **Elephants** can detect **infrasound**.
- A 'silent' dog whistle gives ultrasound. A dog hears it, but we do not.

Loudness:

- Humans perceive (sense and understand through the senses) the **amplitude** of a sound wave as **loudness**.
- Sounds with **larger amplitude** are heard **louder**, while those with **smaller amplitude** sound **softer**.
- Loudness **decreases** as we move farther away from the source, as the intensity decreases.
- In everyday language, loudness and intensity are often used interchangeably (in place of each other). But **intensity** is a **measurable quantity**, whereas **loudness depends on the listener's hearing ability**. The same sound may seem loud to one person and soft to another.

Decibel:

- Loudness of sound is commonly measured in **decibels (dB)**.
- Even a **small increase** in the dB level means a **large increase** in the sound intensity. (An increase of 10 dB means about 10 times more intensity.)
- **Note:** loudness depends on the listener, but the decibel value is measured by an instrument (sound level meter) from the intensity, so it is the same for everyone. That is why the decibel scale is used as a practical, measurable scale of loudness.
- **Examples:** very soft sounds like **rustling leaves** are around a **few dB**; **normal conversation** is about **60 dB**; very loud sounds like **firecrackers** can be **more than 100 dB**.

Noise and hearing:

- Unwanted or harmful sound is called **noise**. **Noise pollution** (harmful level of unwanted sound in the surroundings) is a severe problem.
- Being exposed to sound levels above the recommended limits, especially for long durations, can affect **health, sleep and hearing**.
- Prolonged (lasting for a long time) exposure to loud sound can lead to **hearing loss**. Hearing loss can be tested using **audiograms** (charts that show how well a person hears sounds of different frequencies).
- People with hearing loss may use **hearing aids**. A hearing aid has a **microphone** (picks up sound), an **amplifier** (makes it stronger) and a **speaker** (sends it into the ear). It helps the wearer hear and communicate better.

How we hear: the human ear:

- When sound enters the ear, it makes a thin membrane called the **eardrum** vibrate.
- **Tiny bones** in the middle ear quickly **amplify** (make larger or stronger) these vibrations.
- The **cochlea** (a coiled, snail-shaped part of the inner ear) changes them into **electrical signals** that rush to the **brain**, which perceives them as sound.
- The ear has three parts: the **outer ear**, the **middle ear** and the **inner ear**.

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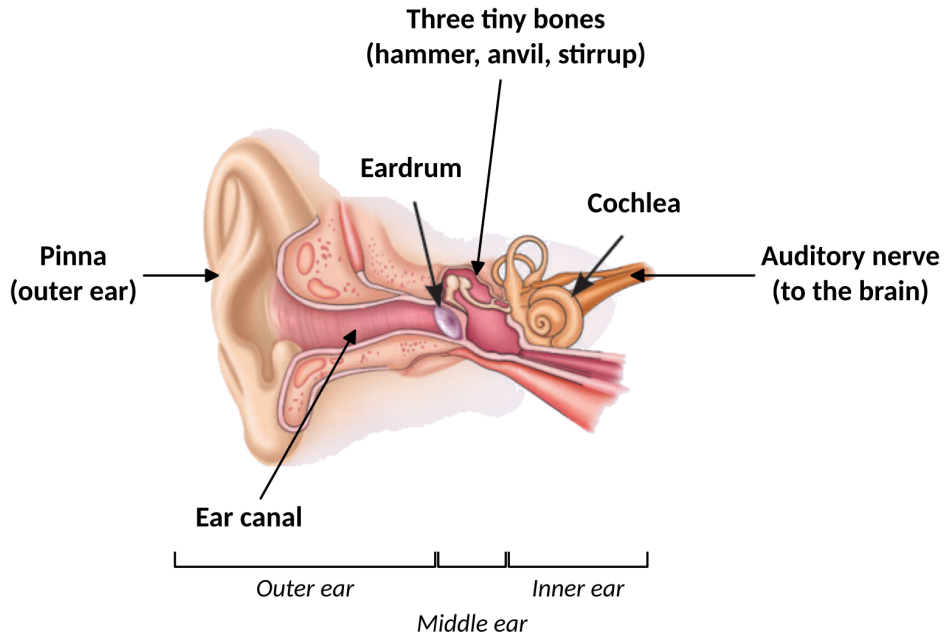


Fig. 21: The human ear with its important parts

Table 2: Important parts of the ear and their functions

Part	Where it is	Function (what it does)
Pinna	Outer ear (the part we see)	Collects sound from the surroundings and sends it into the ear canal.
Ear canal	Outer ear (a tube)	Carries the sound to the eardrum.
Eardrum	At the end of the ear canal (between outer and middle ear)	A thin, tightly stretched membrane. Sound makes it vibrate: a compression pushes it in and a rarefaction lets it move out.
Three tiny bones (hammer, anvil, stirrup)	Middle ear	Pick up the vibrations of the eardrum, amplify (make stronger) them and pass them to the inner ear. They are the smallest bones in our body.
Cochlea	Inner ear (coiled, snail-shaped, filled with liquid)	Changes the vibrations into electrical signals.
Auditory nerve	Joins the cochlea to the brain	Carries the electrical signals to the brain.

- **Path of sound in the ear:** sound → pinna → ear canal → eardrum (vibrates) → three tiny bones (amplify) → cochlea (vibrations changed into electrical signals) → auditory nerve → brain (understands them as sound).
- The eardrum is very thin, so very loud sounds or sharp objects put in the ear can damage it and cause hearing loss.
- Having **two ears** helps the brain find the **direction** from which a sound comes.
- The brain compares which ear hears the sound first. From the tiny time gap between the two ears (often **less than a thousandth of a second**), it finds where the sound came from.
- Animals hear in different ways. **Snakes and fish** sense vibrations through their bodies, while some **insects** have ear-like organs on their body parts (for example, crickets have them on their legs).

Tone and musical note:

- A **tone** is a sound of a **single frequency**, like the sound of a tuning fork or of whistling.
- A **musical note**, like the sound of a plucked **tanpura** (a long-necked Indian string instrument that gives a steady background sound) string or of singing, is a **combination** of frequencies:
  - the **lowest** frequency, called the **fundamental**, and
  - **higher** frequencies, called **overtones**.
- Together they create a **rich and pleasant** sound.
- Indian string instruments like the **sarangi** (a bowed Indian string instrument)**, sitar and veena** (an ancient plucked Indian string instrument) often use **extra strings** to enrich this combination. These extra strings vibrate on their own when the main strings are played.

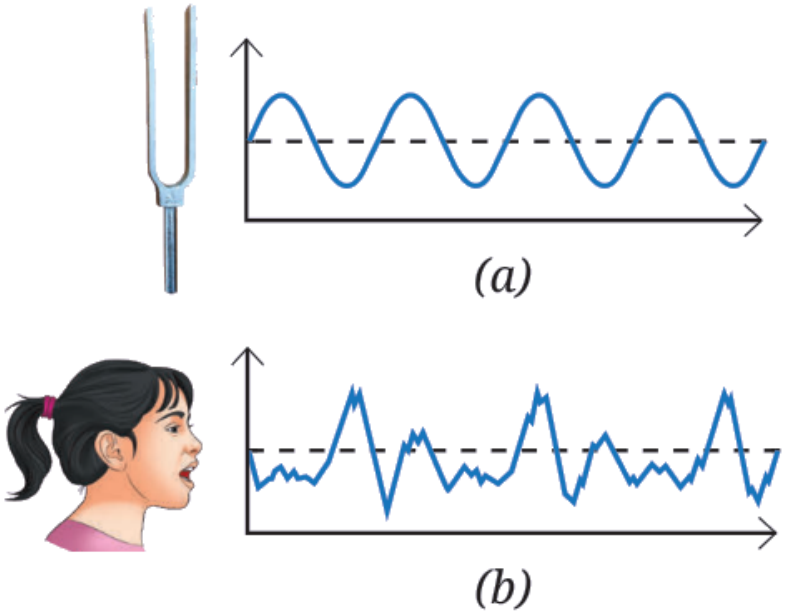


Fig. 22: Graph of the sound of (a) a vibrating tuning fork or a person whistling (a tone) and (b) a child singing (a musical note)

Timbre:

- **Timbre** is the quality of sound that makes each instrument (or voice) sound **unique**, even when the **same note** is played at the **same loudness**.
- It comes from the instrument's **shape, material and construction**, which decide the **pattern and intensity of the overtones**.
- **Note:** two sounds can have the same pitch (same fundamental frequency) and the same loudness, and still differ in timbre, because their overtones are different.
- **Example 1:** a **flute, ektara** (a simple one-string folk instrument) **and tabla** playing the same note at the same loudness each sound different.
- **Example 2:** we can recognise a friend's voice without seeing the friend.

Octave:

- An **octave** is the interval between two notes where one has **double the frequency** of the other.
- It is measured between their **fundamental frequencies**.
- **Example:** notes of **200 Hz and 400 Hz** are one octave apart. The lower 'Sa' and the upper 'Sa' are also one octave apart.

C. V. Raman and the acoustics of Indian drums:

- **Sir C. V. Raman** won **India's first Nobel Prize in Science** (Physics, 1930) for discovering the **Raman Effect** in light.
- He also made important contributions to **acoustics** (the science of sound and how it behaves in a place) by studying Indian **percussion instruments** (instruments that are played by striking, like drums), such as the **tabla** and **mridangam** (a two-sided drum used in South Indian (Carnatic) music).
- He found how these drums produce **rich and nuanced** (having fine and small differences) sounds, unlike most other drums.

Syaahi of the tabla:

- Indian drums like the **tabla** or **mridangam** have a **black patch** at the centre of the drum head membrane, called the **syaahi**.
- This patch **changes the vibration** of the membrane, so these instruments produce a **rich variety of sounds**.
- The syaahi also gives a level of **tonal control** (control over the quality and pitch of the sound produced) rarely found in other drums.

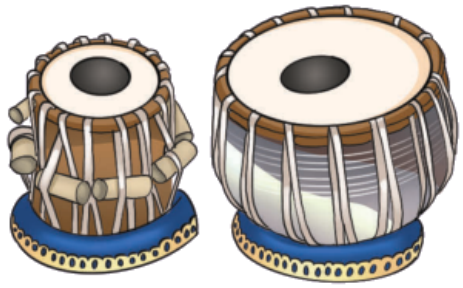


Fig. 23: A tabla set (the black patch at the centre is the syaahi)

Reflection of Sound

➢ Sound waves can **bounce back** from obstacles (things that block the way) like solids or liquids. This is called the **reflection of sound**.

➢ Sound follows the **same laws of reflection as light** (learnt in earlier grades):

- The angle of incidence is equal to the angle of reflection. That is, the incident sound and the reflected sound make **equal angles with the normal** to the reflecting surface at the point of incidence.
- The incident sound, the reflected sound and the normal at the point of incidence all lie in the **same plane**.

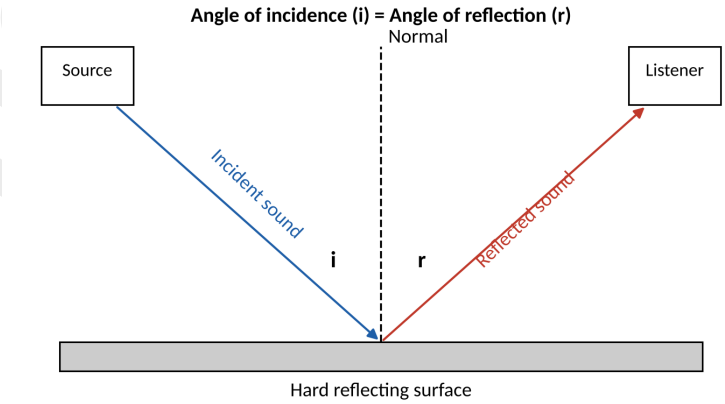


Fig. 24: Reflection of sound from a hard surface (angle i = angle r)

➢ One of the most common examples of reflection of sound is an **echo**.

✓ **Echo:**

- An **echo** is the sound heard again after it is **reflected** from the hard surface of a distant object and comes back to us.
- **Example:** if we shout near a mountain, a cliff or in a long corridor, we may hear our voice again after some time.
- **Why echoes are not heard everywhere:**
  - In a small room, reflections from the walls arrive **too quickly**. The brain cannot separate them from the original sound.
  - If the time gap between two sounds reaching us is **at least 0.1 s**, we hear them as **separate sounds**. If it is **less than 0.1 s**, we cannot clearly tell them apart.
  - The sensation of a sound stays in our brain for about 0.1 s, and that is why this time limit exists.
- **Minimum distance for an echo:**
  - This 0.1 s time gap helps us find the **minimum distance** of a reflecting surface from which an echo can be heard.
  - Taking speed of sound = 340 m s^{-1}, distance travelled in 0.1 s = speed × time = 340 × 0.1 = **34 m**.
  - These 34 m are travelled from the source to the reflecting surface **and back**.
  - So, the **minimum echo distance** = 34 ÷ 2 = **17 m**.

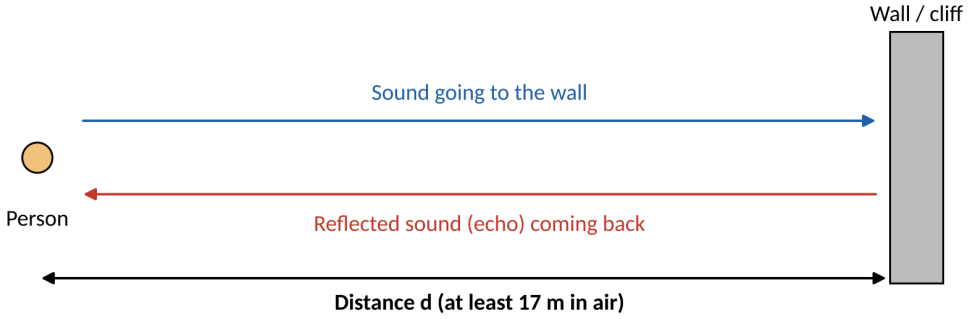


Fig. 25: Minimum distance of a reflecting surface for hearing an echo

- **Surfaces and echoes:**
  - Echoes are stronger from **hard and smooth** surfaces, as these reflect sound well.
  - **Soft** surfaces like curtains **absorb** sound.
  - **Rough** surfaces **scatter** sound in different directions.
  - In such cases (soft or rough surfaces), echoes cannot be heard clearly.
- **Solved Example 5:**
  - **Question:** You clap in an empty corridor and hear an echo after 0.5 s. If the speed of sound in air is 340 m s^{-1}, find your distance from the wall.
  - Sound travels to the wall and back, so the total distance is twice the distance to the wall.
  - Distance from wall = (v × t) ÷ 2 = (340 × 0.5) ÷ 2 = **85 m**

✓ **Reverberation:**

- In a large hall or auditorium, the sound can be **reflected many times** from the walls, ceiling and floor.
- These multiple reflections make the sound **persist** (continue) even after the source stops producing it. This is called **reverberation**.

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- It happens when the sound reflections from surfaces arrive with a time difference of **less than 0.05 s**.
- **Note (gap between 0.05 s and 0.1 s):** if a reflected sound comes back after 0.05 s but before 0.1 s, it is still too early to be heard as a separate echo. It joins the original sound and makes it last a little longer and less clear. So: **0.1 s or more → echo**; **less than 0.1 s → no separate echo**; many reflections arriving **less than 0.05 s apart → reverberation**.
- Modern auditoriums and large concert halls are designed so that they have only the **desirable** amount of reverberation. Then the audience everywhere can hear speech, music and other sounds clearly, without any distortions (unwanted changes that spoil the sound).
- **Sound absorbing panels, upholstered** (covered with soft padding and cloth) **chairs, curtains** and other **soft, porous surfaces** reduce unwanted reverberation, as they absorb sound. Too much reverberation would make the sound **garbled** (mixed up and unclear).

Echo	Reverberation
A single, clear repetition of sound after reflection.	Many reflections that mix together and make the sound last longer.
Reflected sound reaches at least 0.1 s after the original sound.	Reflections reach with a time difference of less than 0.05 s.
Needs a reflecting surface at least 17 m away (in air).	Happens in large closed halls and auditoriums.

- **Gol Gumbaz: the Whispering Gallery:**
  - **Medieval** (belonging to the Middle Ages (about 500 to 1500 CE)) architects across the world, including in India, designed several monuments with a deep understanding of **acoustics**.
  - The famous **Whispering Gallery** of the **Gol Gumbaz** in **Bijapur (Vijayapura), Karnataka** has a remarkable design.
  - Even a **faint whisper** can be heard **many times** across the large **dome** (a round roof shaped like half of a ball), as the sound is reflected again and again by its curved wall.

Ultrasonic and Infrasonic Waves, and their Applications

➢ Sound waves with frequencies **outside the human audible range** have important uses in science, medicine and technology.

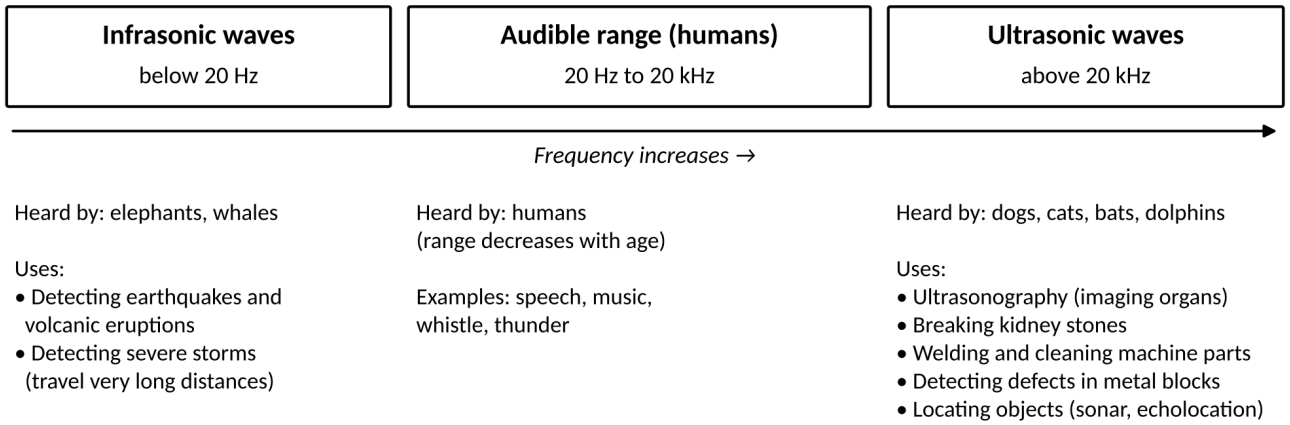


Fig. 26: Ranges of sound waves and their uses

- **Uses of infrasonic waves (less than 20 Hz):**
  - To detect natural events, such as **earthquakes and volcanic eruptions**.
  - To detect **severe storms**, as such waves travel long distances through air and the Earth.
- **Uses of ultrasonic waves (more than 20 kHz):**
  - To make images of **internal organs without surgery**. This is called **ultrasonography**.
  - To break **kidney stones** into smaller pieces, which can then pass out of the body.
  - For **ultrasonic welding** and for **cleaning delicate machine parts** in industries.
  - To detect **defects (cracks) inside metal blocks** during construction and industrial testing.
  - To **locate objects** when the sound waves get reflected from them (echolocation and sonar).

✓ **Echolocation:**

- The ability to **locate objects using reflected sound waves** is called **echolocation**.
- Animals that use it send out sound (mostly ultrasonic), and sense the **echoes** to find the position of **obstacles and prey**.
- **Example 1 (bats):** bats are **nocturnal** (active at night) creatures. They fly and search for their prey in the dark without bumping into objects. Most bats send out **short bursts of ultrasonic waves**, which are reflected from nearby objects, and by sensing the echoes they find obstacles and prey.
- **Example 2: dolphins, whales and some birds** also use echolocation for navigation (finding the way from one place to another) and hunting.

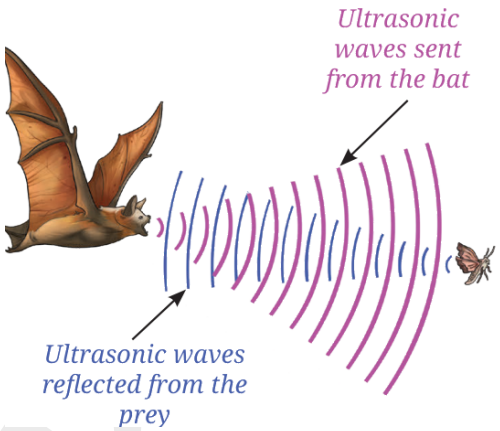


Fig. 27: Echolocation by a bat: ultrasonic waves sent by the bat are reflected from the prey

- **Example: if humans could hear ultrasound like dogs:**
  - **Advantages:** we could hear the calls of bats and other animals, could sense some machine faults early, and could 'hear' ultrasonic sensors.
  - **Disadvantages:** we would be disturbed by many extra sounds from machines, bats, parking sensors, ultrasonic cleaners and pest repellers. It would be harder to rest, sleep and focus.

✓ **Sonar:**

- Humans use the same idea for **underwater exploration** through **sonar** (**SO**und **NA**vigation and **R**anging).
- In sonar, **ultrasonic waves** are sent into water from a ship and the **reflected waves** are studied.
- This gives the **distance, direction and speed** of underwater objects, such as **submarines** (ships that can travel under water) or **shipwrecks** (remains of ships that have sunk).
- Sound travels much farther in water than light, as light gets absorbed by sea water within a short

Conceptual Learning

depth, while sound can go for kilometres. So sound is used for many underwater uses, like finding the **depth of the sea**.

- If t is the time taken by the signal to go and come back and v is the speed of sound in water, then distance of the object = (v × t) ÷ 2.

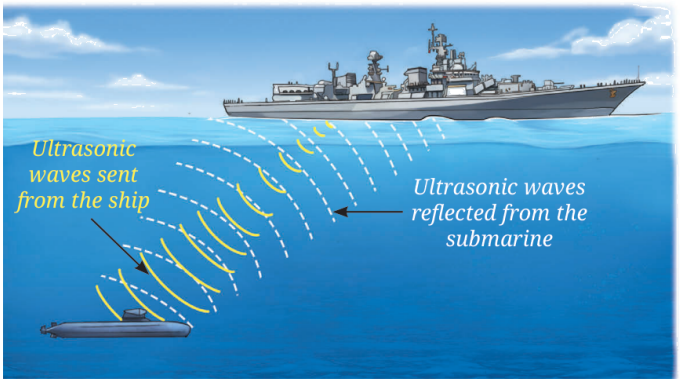


Fig. 28: Working of sonar: ultrasonic waves sent from the ship are reflected from a submarine

- **Solved Example 6:**
  - **Question:** A naval sonar signal sent into seawater returns after 0.90 s. The speed of sound in seawater is 1530 m s^{-1}. How far is the object?
  - Time taken to reach the object and come back = 0.90 s
  - Time taken to reach the object = 0.90 ÷ 2 = 0.45 s
  - Distance = speed × time = 1530 × 0.45 = **688.5 m**

✓ **Audio surveillance:**

- **Drones** and aircraft produce a **characteristic sound** (their own typical sound) from their motors and engines.
- Even when they are hard to see, the **low frequency humming** they produce can be detected by sensitive **sound sensors**.
- This method is called **audio surveillance** (watching by listening). It helps to **monitor the airspace** for safety and security.



Fig. 29: A drone

✓ **New uses of sound:**

- Sound helps us explore places and events beyond human hearing.
- **Space probes** (spacecraft without people, sent to study planets and space) have recorded the **first sounds from Mars**.
- Scientists time the sound of distant **earthquakes** travelling through the sea to measure tiny changes in **ocean temperature**. This helps to understand the Earth's changing climate.

Conceptual Learning

- Biologists use the **buzz of mosquitoes** to identify **disease-carrying mosquitoes**.
- Researchers listen to the tiny **crackles** (short, sharp, breaking sounds) produced by **microbes** (very tiny living organisms seen only with a microscope) **in the soil** to study **soil health and biodiversity**.
- As technology improves, sound is becoming an even more powerful tool to explore planets, living organisms and the hidden activities of nature.

Activities

✓ **Activity 1: Let us explore (sound from a vibrating rubber band):**

- **Aim:**
  - To show that sound is produced by vibrating objects.
- **Steps:**
  1) Take a cardboard box with one side open and a rubber band.
  2) Stretch the rubber band across the open side of the box.
  3) Hold the box steady with one hand and pluck the rubber band with a finger. Listen for a sound.
  4) Pluck the rubber band again and watch it carefully to see if it is vibrating.
  5) Wait till the rubber band stops vibrating and check if the sound is still heard.
  6) Change the tension in the rubber band by stretching it more or loosening it slightly, and pluck it each time. Note the change in sound.
  7) Remove the rubber band from the box. Stretch it between two fingers and pluck it near your ear. Compare the loudness with the earlier sound.



Fig. 30: A vibrating rubber band on a box produces sound

- **Observations:**
  - A sound is heard when the rubber band is plucked.
  - The rubber band is seen vibrating (it looks blurred) while the sound is heard.
  - When the rubber band stops vibrating, the sound also stops.
  - When the band is stretched more (more tension), the sound becomes **higher (shriller)**. When it is loosened, the sound becomes **lower (deeper)**, as a tighter band vibrates faster.
  - Without the box, the sound is still produced but it is **much softer**, as the box and the air inside it also vibrate and make the sound louder.
- **Conclusion:**
  - As long as the rubber band is stretched and vibrating, sound is produced. Once the vibration stops, so does the sound.

Conceptual Learning

- So, **sound is produced by vibrations**.

✓ **Activity 2: Let us explore (vibrating tuning fork):**

Aim:

- To show that the prongs of a sounding tuning fork vibrate, and that sound spreads in different directions.

Steps:

1) Take a tuning fork and a soft rubber pad.
2) Hold the tuning fork by its stem.
3) Strike one of the prongs gently against the rubber pad and bring it close to your ear. (Do not strike it against a hard surface.)
4) Gently touch a water surface with one of the vibrating prongs.
5) Repeat step 3 a few times, bringing the prongs near your ear in different orientations.

Observations:

- A sound is heard when the vibrating tuning fork is brought near the ear.
- When the prong touches water, small waves form on the water surface, and drops may splash out.
- The sound is heard in every orientation of the tuning fork near the ear.



(a) Striking a tuning fork on a rubber pad



(b) A vibrating prong touching water

Fig. 31: Showing that the prongs of a sounding tuning fork vibrate

Conclusion:

- The waves on the water show that the prongs of the tuning fork are **really vibrating**. This supports the idea that **sound is produced by vibrating objects**.
- As the sound is heard in all orientations, sound **propagates in many directions** from the source.

✓ **Activity 3: Let us investigate (sound through a solid):**

Aim:

- To find out whether sound can travel through solids.

Steps:

1) You and your friend stand on opposite sides of a desk in the classroom.
2) Your friend gently knocks or scratches the desk. Listen carefully with your ear in the air.
3) Now place one ear against the desk, close your other ear and listen again.

Observations:

- With the ear in the air, the gentle knock or scratch is heard faintly or not at all.
- With the ear against the desk, the same sound is heard **clearly and louder**.

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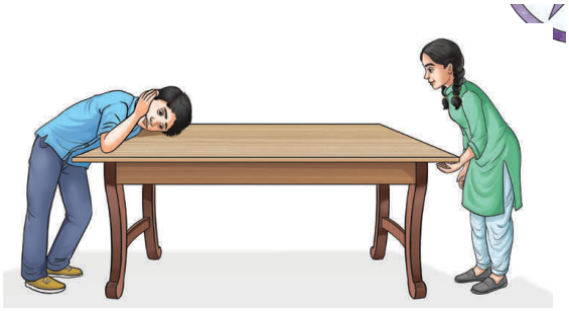


Fig. 32: Listening to a scratch through a desk

- **Conclusion:**
  - **Sound can travel through solids**. It travels better through the wooden desk than through air, as the particles of a solid are closely packed.

✓ **Activity 4: Let us investigate (sound through a liquid):**

- **Aim:**
  - To find out whether sound can travel through liquids.
- **Steps:**
  1) Take a large tub or bucket filled with water up to the brim (top edge of a container) and two metal spoons.
  2) Tap the spoons against each other in air and listen to the sound.
  3) Now submerge (put fully under water) the two spoons in water, without touching the sides or bottom of the bucket, and tap them against each other again.



(a) In air *(b) Under water*

Fig. 33: Tapping two spoons against each other in air and in water

- **Observations:**
  - A sound is heard when the spoons are tapped in air.
  - A sound is **also heard** when the spoons are tapped under water.
- **Conclusion:**
  - The sound of the submerged spoons reaches us after travelling through **water and then air**. If sound did not travel through liquids, we would not have heard it.
  - So, **sound can travel through liquids**. Together with Activity 3, this shows that sound travels through solids, liquids and gases.

✓ **Activity 5: Let us observe (disturbance in a slinky):**

- **Aim:**
  - To understand, using a slinky, how a disturbance travels through a medium while the parts of the

Conceptual Learning

medium only oscillate.

- **Steps:**
  1) Take a slinky and a marker.
  2) Make a mark on one turn of the slinky. Lay the slinky horizontally on a table or the floor.
  3) Ask a friend to hold one end of the slinky fixed, while you hold the other end, keeping it slightly stretched.
  4) Give your end a sharp push towards your friend and then quickly pull it back. Watch for a disturbance moving towards your friend.
  5) Now push and pull the end many times quickly, one after another (this is like sound being produced continuously). Watch the disturbances and the mark.

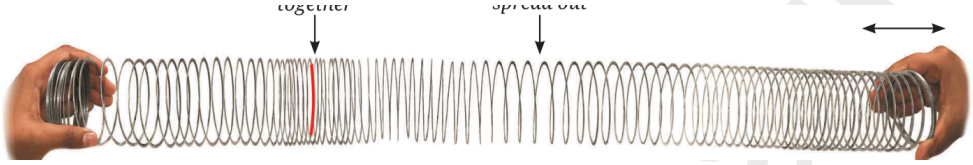


Fig. 34: Disturbance travelling along a slinky: turns closer together and turns more spread out

- **Observations:**
  - A disturbance is created at your end and it moves along the slinky towards your friend.
  - On pushing and pulling many times, a **series of disturbances** is produced and they move across the length of the slinky.
  - There are regions where the turns are **closer together** than usual, and regions where they are **more spread out**. These regions travel along the slinky.
  - The mark **does not travel** along the slinky. It only moves back and forth about its position of rest, **parallel** to the direction of the disturbance.
- **Conclusion:**
  - Each turn oscillates about its own position, while the **disturbance** travels from one end to the other.
  - Sound moves through a medium in the same way. The closer and spread-out regions stand for **compressions** (higher density) and **rarefactions** (lower density) of air. The wave is **longitudinal**.

✓ **Activity 6: Let us experiment (sound moves grains):**

- **Aim:**
  - To show that sound carries energy.
- **Steps:**
  1) Take a wide-mouthed container, a cellophane (a thin, transparent plastic-like sheet used for wrapping) or rubber sheet (like that of a balloon) larger than its opening, a loud sound source (such as a metal plate and a beater) and some grains (rice, semolina, salt or chalk powder).
  2) Stretch the sheet tightly over the mouth of the container and fix it with tape or a rubber band.
  3) Sprinkle the grains evenly over the sheet, so that they are not clumped together.
  4) Produce a loud sound near the container without touching it. Observe the grains.
  5) Repeat with different sources of sound, with louder and softer sounds, and with different grains.
- **Observations:**
  - The grains on the sheet **move and jump**, even though the source of sound does not touch the

Conceptual Learning

sheet or the container.

- When the sound is **louder** (plate struck harder), the grains **jump higher**. With a softer sound, they move less.
- Lighter grains (like semolina or chalk powder) move more easily than heavier ones.

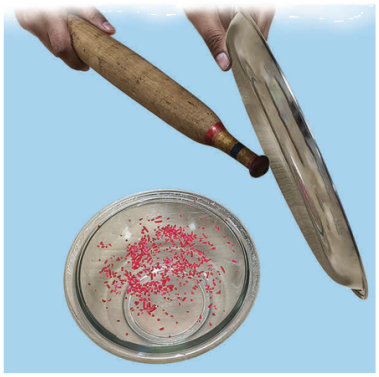


Fig. 35: Grains on a stretched sheet jump when a loud sound is produced nearby

Conclusion:

- Sound travels through air, reaches the sheet and makes it vibrate. This vibration makes the grains move.
- So, **sound is a form of energy**. A vibrating source transfers energy to the medium, and the energy is carried forward by the sound wave.
- A louder sound (larger amplitude) carries **more energy**, so the grains jump higher.

✓ **Activity 7: Let us experiment (frequency of musical notes):**

Aim:

- To find the frequencies of the musical notes Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa and compare them. (A demonstration activity done in class with the teacher.)

Steps:

1) Use a mobile app, such as Phyphox, that can find the frequencies of sounds. Use its 'Audio Spectrum' option, which shows the frequency as a graph or in hertz (Hz).
2) Sing the musical notes 'Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa' one after another, or play them with a music or tone app on another phone. Watch how the frequency changes with each note.
3) Record the approximate frequency of each musical note.
4) Compare the notes by finding the ratio of each frequency to that of 'Sa'. Look for a pattern.
5) If both voice and mobile-generated notes are used, compare their frequencies for the same notes.

Observations:

- The frequency is **lowest for 'Sa'** and gradually **increases** for Re, Ga, Ma, Pa, Dha and Ni.
- Sample readings (when Sa is taken as 240 Hz): Sa 240, Re 270, Ga 300, Ma 320, Pa 360, Dha 400, Ni 450, upper Sa 480 Hz.
- Ratios with Sa: 1, 9/8, 5/4, 4/3, 3/2, 5/3, 15/8 and 2. The upper 'Sa' has **double** the frequency of the lower 'Sa' (one octave).
- The voice and the app give nearly the same frequency for the same note, but the voice values change a little each time.

Conceptual Learning

- **Conclusion:**
  - Each musical note has a **distinct frequency**, which makes it sound different.
  - The ratios of the frequencies of the notes to 'Sa' follow a fixed pattern, whatever the starting frequency of 'Sa'.

✓ **Activity 8: Let us experiment (sounds of different frequencies):**

- **Aim:**
  - To listen to sounds of different frequencies and find the lower limit of hearing. (A demonstration activity done in class with the teacher.)
- **Steps:**
  1) Open a mobile app that can generate sounds.
  2) Set the frequency to 100 Hz, tap 'play' and listen carefully.
  3) Increase the frequency in steps of 100 Hz up to 1000 Hz and note how the sound changes.
  4) Next, set the frequency to 50 Hz. Reduce it till about 20 Hz, or till the point where the sound can no longer be heard.
- **Observations:**
  - As the frequency increases from 100 Hz to 1000 Hz, the sound becomes **higher in pitch (shriller)**.
  - As the frequency is reduced from 50 Hz, the sound becomes **deeper** and harder to hear. Near 20 Hz it is almost not heard at all (it may be felt more as a vibration).
  - The lowest frequency that can be heard is a little different for different students.
- **Conclusion:**
  - Higher frequency is heard as a **higher pitch** and lower frequency as a **lower pitch**.
  - Humans can hear only a **limited range** of frequencies. The lower limit is about **20 Hz** (the upper limit is about 20 kHz).

Conceptual Learning